# Department of CSE(AIML), PES University
# UE26CS151A -Python for Computational Problem Solving
# Laboratory Week 6

**Note:** Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Problem Statements

**Q1. Count the Number of Lines Containing a Given Word** Write a program to read a text file and count the number of lines containing a word entered by the user.

- **User Input:** File name (string) and search word (string) entered by the user.
- **Data Types:** Use string for words and int for line count.

**Q2. Find Students Who Scored Above a Given Mark Using a Dictionary** Write a program to store student names and marks in a dictionary and display the names of students whose marks are greater than a user-entered value.

- **User Input:** Student names (string), marks (int), and threshold mark (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

**Q3. Find the Total Quantity Sold for Each Product from a CSV File** Write a program to read product names and quantities sold from **week_6_set4_q3.csv** and use a dictionary to calculate and display the total quantity sold for each product.

- **User Input:** No user input required.
- **Data Types:** Use dict with string keys (product names) and int values (quantities).
- **File Reference:** Refer to **week_6_set4_q3.csv** for input.

**Q4. Display Unique Words from a Text File** Write a program to read a text file and display each word only once, preserving its first occurrence order.

- **User Input:** File name entered by the user.
- **Data Types:** Use list for storing words and set for uniqueness check.

**Q5. Find the Highest-Paid Employee from a CSV File** Write a program to read employee names and salaries from **week_6_set4_q5.csv** and display the name and salary of the highest-paid employee.

- **User Input:** No user input required.
- **Data Types:** Use string for employee names and int for salaries.
- **File Reference:** Refer to **week_6_set4_q5.csv** for input.

**Q6. Create a Dictionary of Subject-wise Marks** Write a program to store subject names and marks in a dictionary. Allow the user to add a new subject, modify an existing subject's marks, and display the final dictionary.

- **User Input:** Subject name (string) and marks (int) entered by the user.
- **Data Types:** Use dict with string keys and int values.

**Q7. Copy Lines Containing Numbers into Another File** Write a program to read a text file and copy only those lines containing at least one digit into another text file.

- **User Input:** Input file name (string) and output file name (string) entered by the user.
- **Data Types:** Use string for lines.

- **File Handling:** Use text file read mode (r) and write mode (w).

**Q8. Write a Python program that asks the user to enter the names of four products and their prices. Store the data in a list of lists and write the results into a CSV file named week_6_set4_q8.csv.**

- **Data Types: string for product names, float for prices.**

Prepared By,
**Prof. Adithi S**
CSE Dept RR Campus